

23 An antibiotic enters bacterial cells through a membrane channel protein, P.

Some bacterial cells have shown resistance to this antibiotic by acquiring a mutation which alters P. This mutation prevents the entry of the antibiotic into the cell.

Which conclusions can be drawn about how resistance to this antibiotic developed in these bacteria?

- 1 The mutation changed the order of the amino acids in the gene coding for P.
- 2 The mutation resulted in the production of P with an altered tertiary structure.
- 3 The antibiotic is a hydrophobic molecule and so cannot cross the phospholipid bilayer to enter the cell.

A 1, 2 and 3 **B** 1 and 3 only **C** 2 and 3 only **D** 2 only